TEAM 8

Joint Training of QAT & Token Pruning

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TEAM 8

Project

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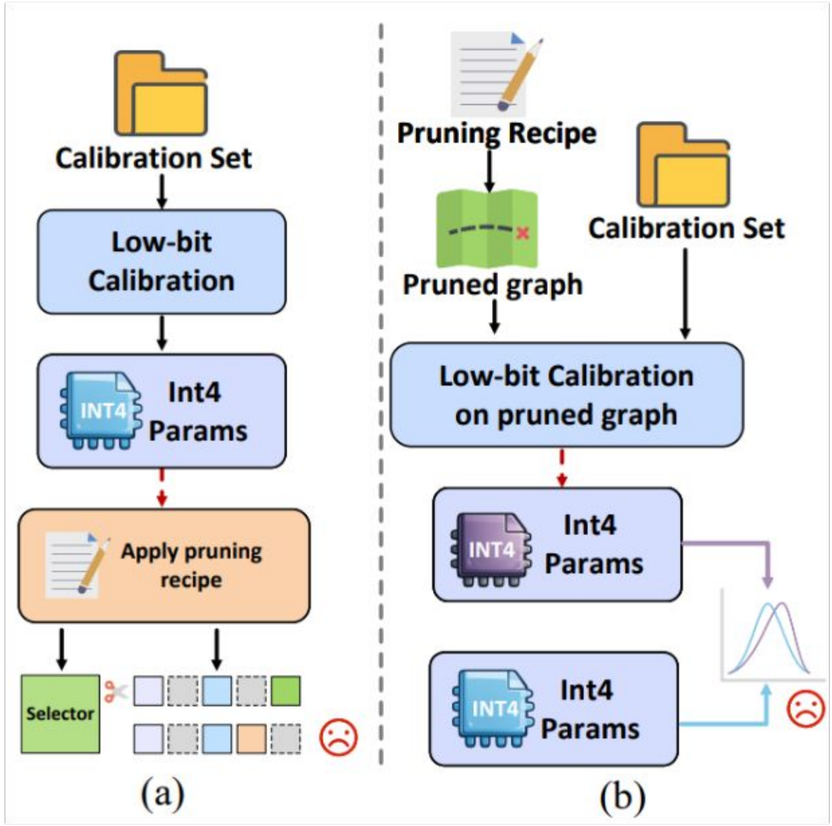
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1.
INTRODUCTION

- motivation
- hypothesis



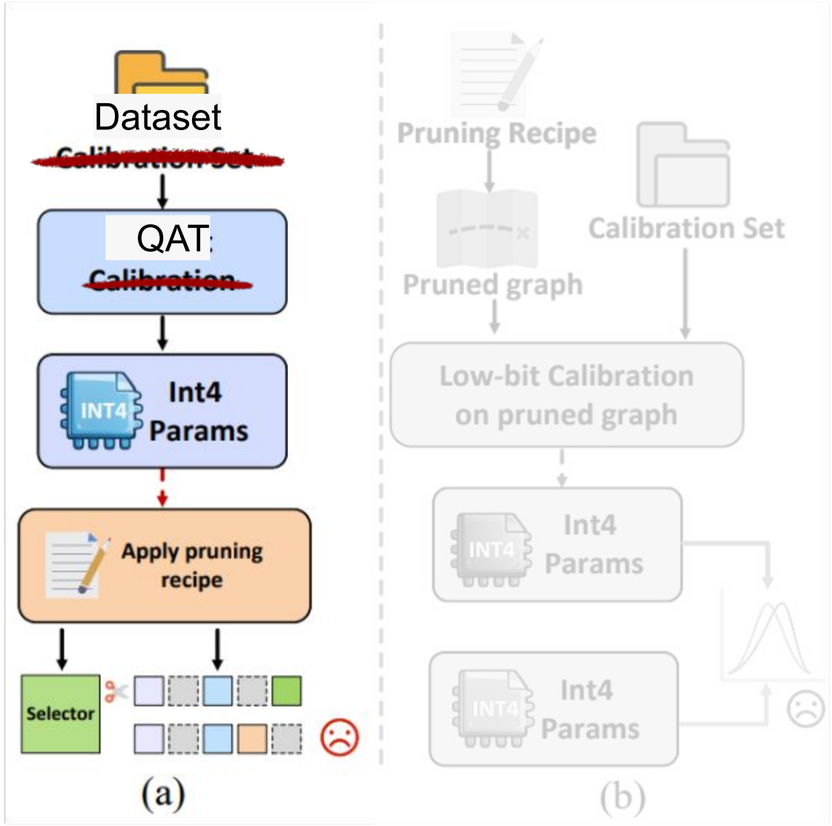
Quantization - Token Pruning Mismatch

a) Quant. 적용 전/후 activation distribution 변화

b) 양자화된 모델의 최적 pruning recipe를 구하지 못 함

Li, Xinqing, et al. "Towards Joint Quantization and Token Pruning of Vision-Language Models." arXiv preprint arXiv:2604.17320 (2026).

token pruning
after QAT



Quantization - Token Pruning Mismatch

QAT 후 배포 시 token pruning을 진행시,
(a)에서 Quant. 적용 전/후 activation distribution 변화 발생

>>> Quantization을 고려한 Token Pruning 방식 필요

Li, Xinqing, et al. "Towards Joint Quantization and Token Pruning of Vision-Language Models." arXiv preprint arXiv:2604.17320 (2026).

QAT 학습 과정에서 weight와 token pruning을 동시에 최적화

기존 방법론

activation distribution이 변화하여
학습 때와 추론 때의 양자화가
깨짐

동시 최적화

Quantization과 token pruning mismatching
완화

2.
RELATED WORKS

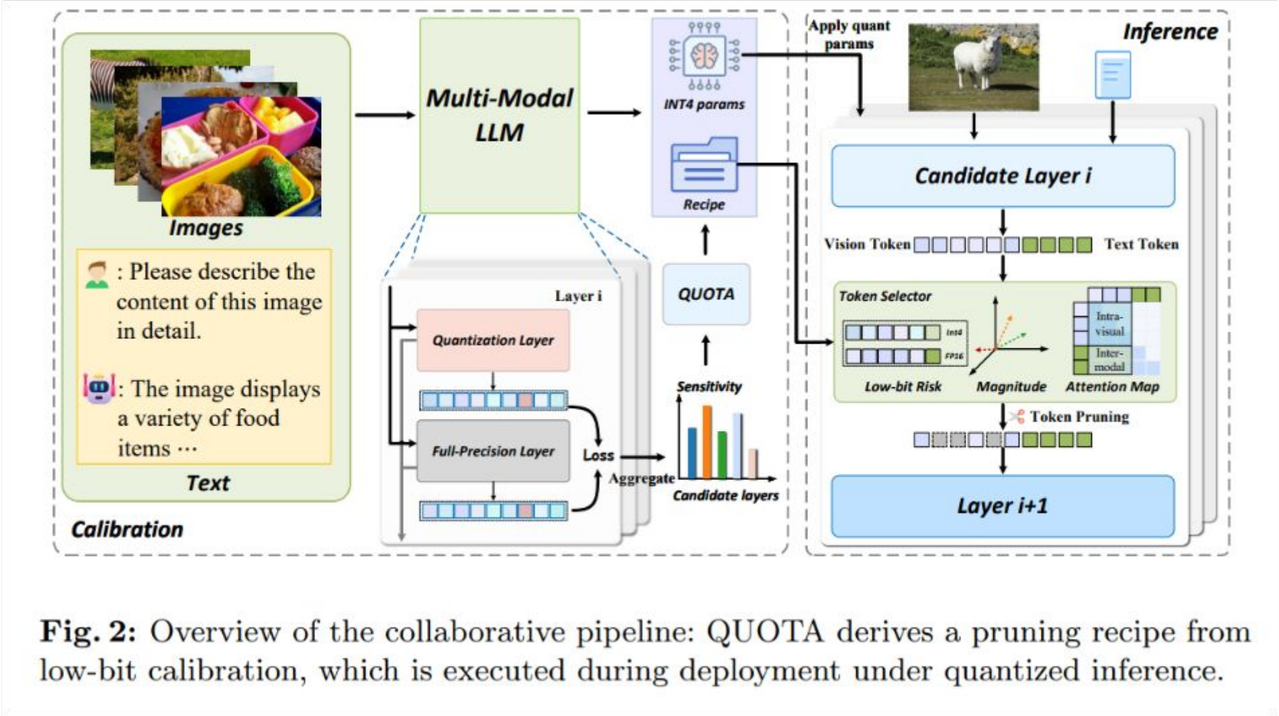


Fig. 2: Overview of the collaborative pipeline: QUOTA derives a pruning recipe from low-bit calibration, which is executed during deployment under quantized inference.

QUOTA

PTQ에 적합한 Token pruning 방법 제안

Quantization Sensitivity 를 추가로 고려하여 pruning 진행

Li, Xinqing, et al. "Towards Joint Quantization and Token Pruning of Vision-Language Models." arXiv preprint arXiv:2604.17320 (2026).

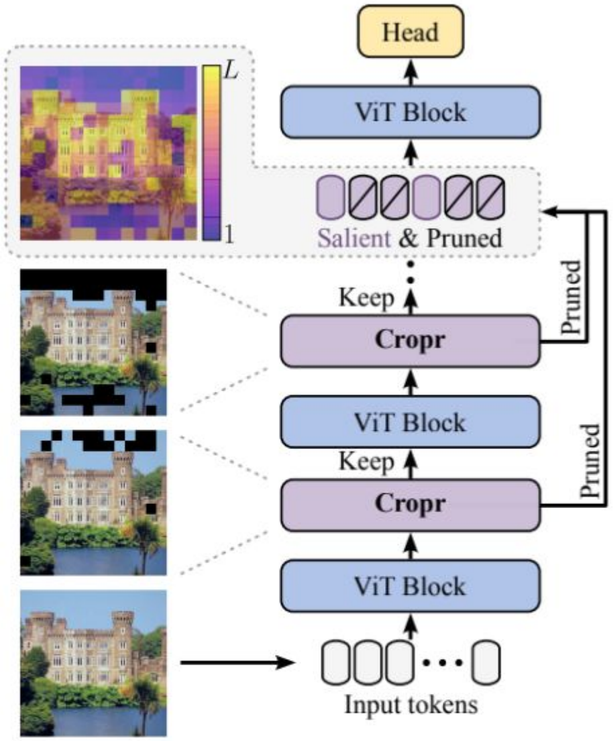
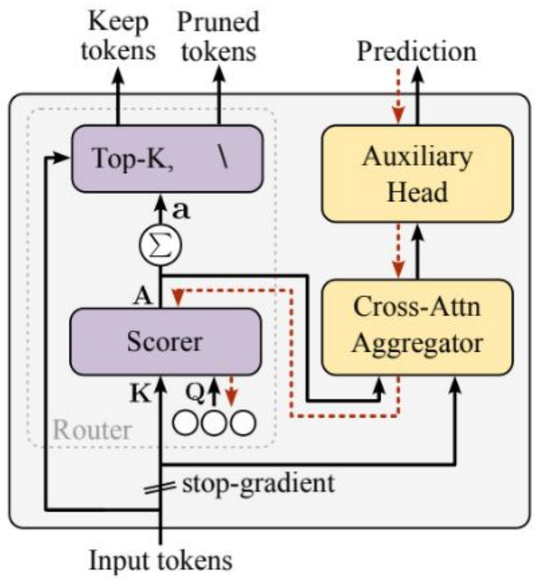


Figure 1. **Cro**ss-attention **pr**uning (**Cropr**) modules successively prune less relevant tokens, retaining only the most discriminative ones for deeper layers. Our method accelerates ViTs while maintaining high performance and is applicable to many vision tasks, from classification to segmentation and detection. The example castle images illustrate the pruning process. The heatmap visualizes which tokens were pruned at each block 1 to L in the network.



Token Cropr

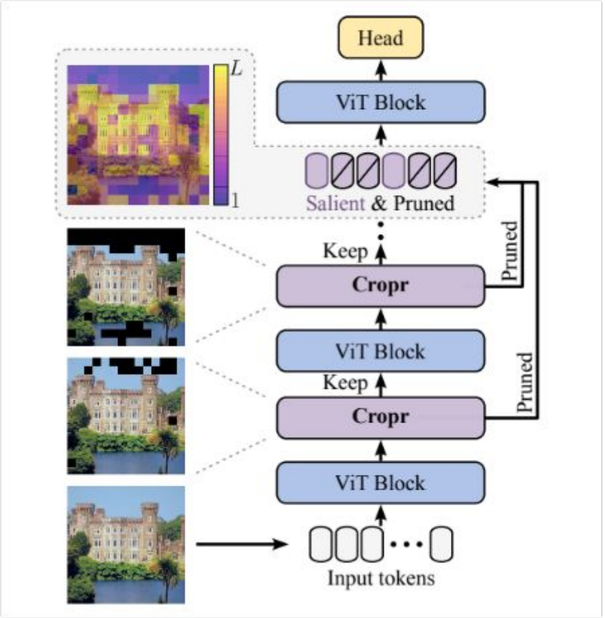
ViT에 token pruning module을 추가하여 학습
Attention score 기반 상위 k개 token 선택

성능 손실 거의 없이 1.5 ~ 4배 속도 향상 달성

Bergner, Benjamin, Christoph Lippert, and Aravindh Mahendran. "Token cropr: Faster vits for quite a few tasks."
2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). IEEE, 2025.

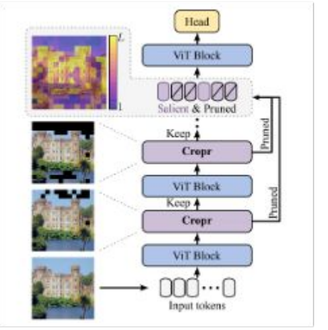
3.
METHOD

- modeling
- formulation

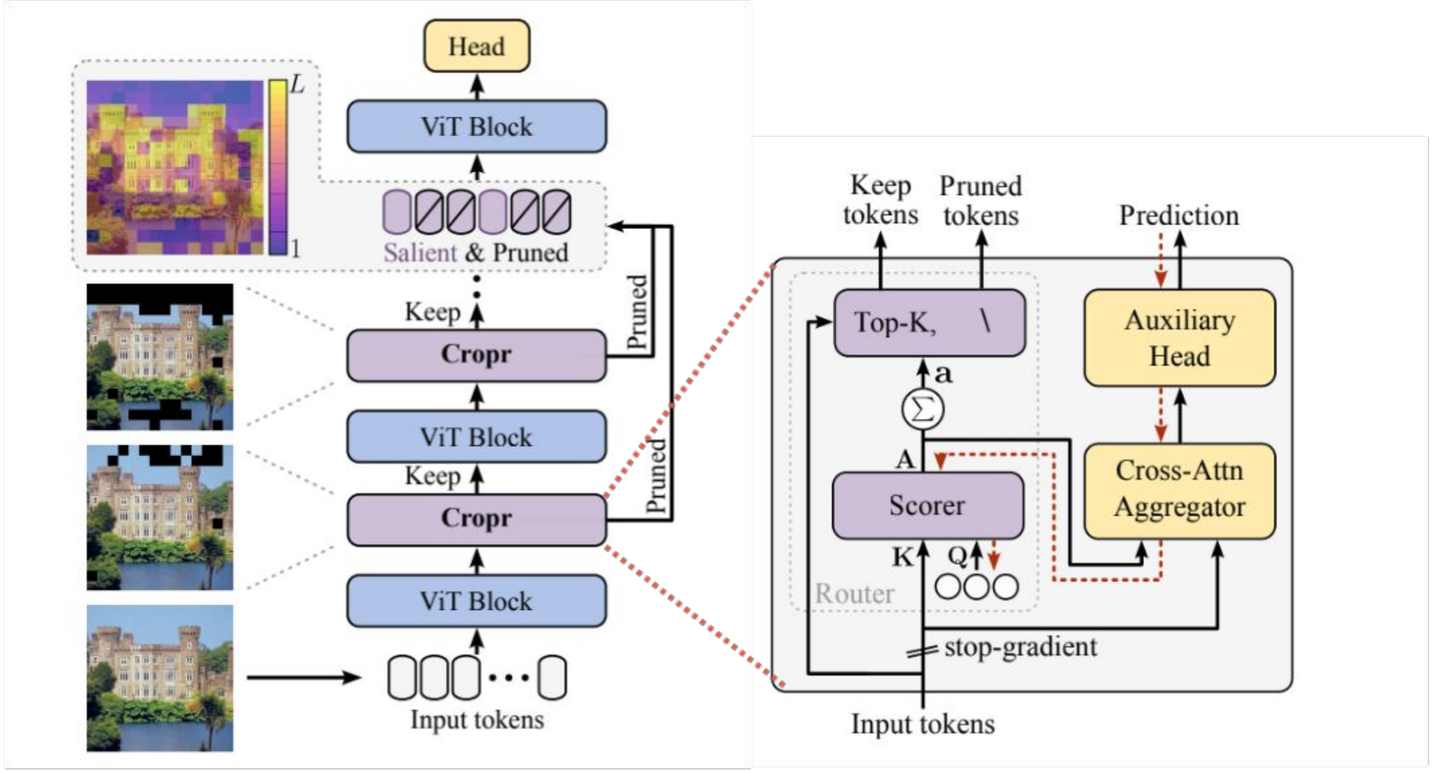


Base Cropr ViT

QAT



Quantized Cropr ViT



$\mathbf{X} \in \mathbb{R}^{M \times D}$ M개의 input tokens

$\mathbf{K} \in \mathbb{R}^{M \times D}$ key matrix, Identity

$\mathbf{Q} \in \mathbb{R}^{N \times D}$ learnable queries

$$\mathbf{X}^k = \text{Top-K}(\mathbf{X} \mid \mathbf{a})$$
$$\mathbf{X}^p = \mathbf{X} \setminus \mathbf{X}^k$$
$$\mathbf{A} = \mathbf{Q}\mathbf{K}(\mathbf{X})^\top = \mathbf{Q}\mathbf{X}^\top$$
$$\mathbf{a} = \sum_{n=1}^{N} \mathbf{A}_n, \quad \mathbf{A}_n \in \mathbb{R}^M$$
$$\mathbf{X}' = \text{softmax}\left(\frac{\mathbf{A}}{\sqrt{D}}\right)\mathbf{X}$$
$$\text{aggregator}(\mathbf{X} \mid \mathbf{A}) = \text{MLP}(\text{LN}(\mathbf{X}')) + \mathbf{X}'$$

QAT와 token pruning간 joint optimization 수식

정의

$$\theta^* = \arg\min_{\theta} L\left(y, f_{\theta}^{Q,P_0}(x)\right)$$

만약 pruning module P가 learnable이라면,

$$\theta^*, \theta_{\mathrm{p}}^* = \arg\min_{\theta,\theta_{\mathrm{p}}} L\left(y, f_{\theta}^{Q,P_{\theta_{\mathrm{p}}}}(x)\right)$$

Q	quantization function
P	pruning module
f	neural network
L	loss function
y	label

Loss Function in Croper ViT

$$L = L_{\text{task}}\left(y, p^{Q}_{\text{pruned}}\right) + \lambda_{\text{aux}} \sum_{\ell} L_{\text{task}}\left(y, p^{Q}_{\text{aux}}\right)$$

$f_{\theta}^{Q, P_{\theta_{\text{p}}}}(x)$

block index

i-th aux head prediction

4.
EXPERIMENTS

- experiments setting
- pruning strategy comparison

QAT와 token pruning을 동시에 최적화하면 정확도 손실이 감소할까?



Base model

timm/vit_tiny_patch16_224.augreg_in21k_ft_in1k
+
Token pruner: Cropr



Dataset

Food-101
(class 101개, 전체 샘플 75,750개)

input size - 224x224



Metrics

- top-1 acc
- GFLOPs
- Params
- Storage cost
- Token pruning ratio
- Latency



Enviroment

NVIDIA RTX 4090

	Learning method	pruning *in* leaning	pruning *in* inference	purpose
QAT only	QAT	-	-	QAT의 정확도 기준선
pruning after QAT	QAT	-	[CLS] attn. top-k	pruning 효과
pruning with QAT	QAT	[CLS] attn. top-k	[CLS] attn. top-k	pruning 적응 효과

	Learning method	pruning *in* leaning	pruning *in* inference	purpose
pruning after PTQ	FP32 finetuning	Cropr	Cropr	pruning 후 quant 성능
pruning with QAT	QAT	Cropr	Cropr	pruning + quant. 공동 최적화 성능



Q&A

Thank you